

2.4 Mori's Cone Theorem

Revised excerpt, corresponding to original pp. 146–157. The ground field k is algebraically closed of characteristic zero throughout. All un-subscripted varieties and curves remain over this field.

Recall that in Chapter 1 we introduced some concepts like cones, extremal rays and the nef cone. For a projective variety X , the cone of effective curves is a subcone of the $\mathbb{R}$ -vector space $N_1(X)_{\mathbb{R}}$. Related to the perfect pairing

$$N^1(X)_{\mathbb{R}} \times N_1(X)_{\mathbb{R}} \longrightarrow \mathbb{R},$$

we have

$$\begin{aligned} \mathrm{NE}_{\mathbb{Q}}(X) &= \left\{ \sum_i a_i [C_i] \mid C_i \subseteq X \text{ an irreducible curve, } a_i \in \mathbb{Q}_{\geq 0} \right\}, \\ \mathrm{NE}(X) &= \left\{ \sum_i a_i [C_i] \mid C_i \subseteq X \text{ an irreducible curve, } a_i \in \mathbb{R}_{\geq 0} \right\}, \\ \overline{\mathrm{NE}}(X) &= \text{the closure of } \mathrm{NE}(X) \text{ in } N_1(X)_{\mathbb{R}}. \end{aligned}$$

Here every sum is finite. The nef cone lies in $N^1(X)_{\mathbb{R}}$ and is the dual cone of $\overline{\mathrm{NE}}(X)$. Recall also that for a Cartier divisor D ,

$$D_{\geq 0} := \{\gamma \in N_1(X)_{\mathbb{R}} \mid (D \cdot \gamma) \geq 0\}, \quad D^{\perp} := \{\gamma \in N_1(X)_{\mathbb{R}} \mid (D \cdot \gamma) = 0\}.$$

We use the notation

$$\overline{\mathrm{NE}}(X)_{D \geq 0} := \overline{\mathrm{NE}}(X) \cap D_{\geq 0},$$

and similarly for > 0 , ≤ 0 and < 0 , also for $\mathbb{R}$ -Cartier divisors. Now we are ready to state the first result of Mori for smooth varieties of arbitrary dimension.

2.4A Cone Theorem in the Nonsingular Case

Theorem 2.4.1 (Cone Theorem). *Let X be a nonsingular projective variety such that K_X fails to be nef.*

- (1) *There are at most countably many rational curves $C_i \subseteq X$ such that $0 < -(C_i \cdot K_X) \leq \dim X + 1$, and*

$$\overline{\mathrm{NE}}(X) = \overline{\mathrm{NE}}(X)_{K_X \geq 0} + \sum_i \mathbb{R}_{\geq 0}[C_i].$$

- (2) *For any $\varepsilon > 0$ and any ample Cartier divisor H , we have*

$$\overline{\mathrm{NE}}(X) = \overline{\mathrm{NE}}(X)_{(K_X + \varepsilon H) \geq 0} + \sum_{i \in I(H, \varepsilon)} \mathbb{R}_{\geq 0}[C_i],$$

where $I(H, \varepsilon) = \{i \mid (C_i \cdot (K_X + \varepsilon H)) < 0\}$ is finite.

Proof. From Grothendieck's Theorem 2.1.11 we know that the scheme $\text{Mor}(\mathbb{P}_k^1, X)$ is the disjoint union of countably many schemes of finite type over k , hence has at most countably many irreducible components. Pick a component B , with its reduced structure, and consider the flat projective family

$$\pi : \mathbb{P}_k^1 \times B \longrightarrow B$$

and the universal evaluation map

$$f^{\text{univ}} : \mathbb{P}_k^1 \times \text{Mor}(\mathbb{P}_k^1, X) \longrightarrow X.$$

For any Cartier divisor D on X , the pullback of $\mathcal{O}_X(D)$ is a line bundle on $\mathbb{P}_k^1 \times B$ and is flat over B . Thus the function

$$b \longmapsto \chi(\mathbb{P}_b^1, f_b^* \mathcal{O}_X(D))$$

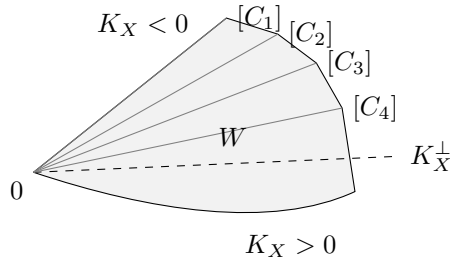
is locally constant. Since B is irreducible, it is connected, so this function is constant. By Riemann–Roch on $\mathbb{P}^1$, the intersection number $((f_b)_*[\mathbb{P}^1] \cdot D)$ is constant. Hence the *pushforward 1-cycles* parametrized by B are numerically equivalent. Every rational curve has a normalization map $\mathbb{P}^1 \rightarrow X$ whose push-forward is its curve cycle, so the numerical classes of rational curves also form an at most countable set.

In each numerical class γ represented by a rational curve with

$$0 < -(\gamma \cdot K_X) \leq \dim X + 1,$$

we pick one such curve C_i . This gives an at most countable index set I , and we define

$$W := \overline{\left(\overline{\text{NE}}(X)_{K_X \geq 0} + \sum_{i \in I} \mathbb{R}_{\geq 0} [C_i] \right)}.$$



Step 1: We aim to prove $W = \overline{\text{NE}}(X)$. Assume the contrary, so the natural inclusion $W \subseteq \overline{\text{NE}}(X)$ is strict. By separation of convex cones, followed by a small ample perturbation and rational approximation, we may choose a Cartier divisor D whose associated linear function $(D \cdot -)$ is positive on $W - \{0\}$ but negative somewhere in $\overline{\text{NE}}(X)$. Indeed, strict positivity can be checked on a compact slice

cut out by an ample divisor, so it survives a sufficiently small perturbation; we then clear the denominators.

Let H be an ample Cartier divisor, and let $\mu > 0$ be the largest real number such that $H + \mu D$ is nef. This number is finite because D is negative somewhere in $\overline{\text{NE}}(X)$. The class $H + \mu D$ lies on the boundary of the nef cone. Thus, by Kleiman's criterion, there is a nonzero $z \in \overline{\text{NE}}(X)$ with

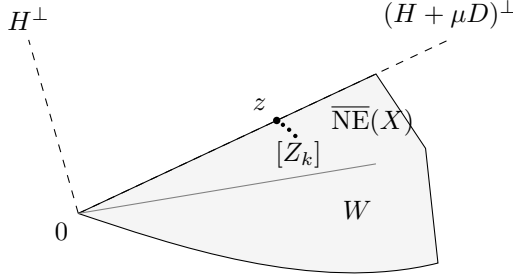
$$((H + \mu D) \cdot z) = 0.$$

Then $(D \cdot z) < 0$ because $(H \cdot z) > 0$, so $z \notin W$. Moreover $(K_X \cdot z) < 0$, since $\overline{\text{NE}}(X)_{K_X \geq 0} \subseteq W$.

Choose effective 1-cycles with rational coefficients $Z_k = \sum_j a_{kj} [Z_{kj}]$, where $a_{kj} > 0$, such that

$$Z_k \longrightarrow z \quad \text{as } k \longrightarrow \infty.$$

The situation is illustrated schematically below.



Fix a rational number $0 < \mu' < \mu$. The $\mathbb{Q}$ -Cartier divisor

$$L := H + \mu' D = \left(1 - \frac{\mu'}{\mu}\right) H + \frac{\mu'}{\mu} (H + \mu D)$$

is ample, being the sum of an ample and a nef $\mathbb{R}$ -divisor. For k sufficiently large, $-(Z_k \cdot K_X) > 0$. By Lemma 2.3.8,

$$\max_j \frac{-(Z_{kj} \cdot K_X)}{(Z_{kj} \cdot L)} \geq \frac{-(Z_k \cdot K_X)}{(Z_k \cdot L)} > 0.$$

Rearrange the indices so that the maximum is achieved for Z_{k0} . Apply Theorem 2.3.7 to Z_{k0} and L : formally, apply it to a positive Cartier multiple of L , then cancel that multiple in the ratio. We obtain a rational curve E_k such that

$$0 < -(E_k \cdot K_X) \leq \dim X + 1 \tag{2.2}$$

and

$$\frac{-(E_k \cdot K_X)}{(E_k \cdot L)} \geq \frac{-(Z_{k0} \cdot K_X)}{(Z_{k0} \cdot L)} \geq \frac{-(Z_k \cdot K_X)}{(Z_k \cdot L)}.$$

By (2.2), E_k is numerically equivalent to one of the curves C_i . Thus $(E_k \cdot D) > 0$, by the choice of D , and consequently

$$\frac{-(E_k \cdot K_X)}{(E_k \cdot H)} \geq \frac{-(E_k \cdot K_X)}{(E_k \cdot L)} \geq \frac{-(Z_k \cdot K_X)}{(Z_k \cdot L)}.$$

On the other hand, there is a positive constant $c \gg 0$ such that $K_X + cH$ is ample. Hence, for every such curve,

$$\frac{-(E_k \cdot K_X)}{(E_k \cdot H)} < c.$$

But $-(z \cdot K_X) > 0$ and $((H + \mu D) \cdot z) = 0$. We can first choose a rational $\mu' < \mu$ sufficiently close to μ that

$$\frac{-(z \cdot K_X)}{(z \cdot (H + \mu' D))} > c.$$

Keep this μ' fixed. Since $Z_k \rightarrow z$, the same strict inequality holds with Z_k in place of z for k sufficiently large, a contradiction. This concludes Step 1.

Step 2: We prove part (2) of the theorem. Fix H and $\varepsilon > 0$. If $(C_i \cdot (K_X + \varepsilon H)) < 0$, then

$$(C_i \cdot H) < \frac{-(C_i \cdot K_X)}{\varepsilon} \leq \frac{\dim X + 1}{\varepsilon}.$$

By the second statement of Theorem 1.4.28, there are only finitely many such numerical classes. Since we chose one C_i per numerical class, there are only finitely many such C_i . Set

$$V_{H,\varepsilon} := \overline{\text{NE}}(X)_{(K_X + \varepsilon H) \geq 0} + \sum_{i \in I(H,\varepsilon)} \mathbb{R}_{\geq 0}[C_i].$$

This cone is closed. Indeed, in a convergent sequence $u_m + \sum_i a_{mi}[C_i]$ in this cone, with $a_{mi} \geq 0$, intersection with H bounds $H \cdot u_m$ and each a_{mi} . The set $\{u \in \overline{\text{NE}}(X) \mid H \cdot u \leq M\}$ is compact for every M . We can therefore pass to a subsequence on which all these terms converge, and the limit remains in $V_{H,\varepsilon}$.

Clearly $V_{H,\varepsilon} \subseteq \overline{\text{NE}}(X)$. It also contains $\overline{\text{NE}}(X)_{K_X \geq 0}$ and every ray $\mathbb{R}_{\geq 0}[C_i]$, and hence, being closed, it contains W . By Step 1, $W = \overline{\text{NE}}(X)$, which proves part (2).

Step 3: We prove part (1), including the removal of the closure in the definition of W . Recall the elementary finite-dimensional convexity fact that a closed convex cone with a compact base is generated by its extremal rays, using finite non-negative sums. To see its applicability here, intersect $\overline{\text{NE}}(X)$ with $H \cdot z = 1$. This is a compact convex set; the finite-dimensional Minkowski theorem and Carathéodory's theorem express every point as a finite convex combination of extreme points. These correspond to extremal rays of $\overline{\text{NE}}(X)$.

Let $R = \mathbb{R}_{\geq 0}r$ be a K_X -negative extremal ray. Choose $\varepsilon > 0$ small enough that $(K_X + \varepsilon H) \cdot r < 0$. By Step 2,

$$r = u + \sum_{i \in I(H, \varepsilon)} a_i [C_i], \quad u \in \overline{\text{NE}}(X)_{(K_X + \varepsilon H) \geq 0}, \quad a_i \geq 0.$$

Extremality implies that every nonzero summand belongs to R . Since $R - \{0\}$ is $(K_X + \varepsilon H)$ -negative, we must have $u = 0$. Thus some C_i generates R . Every remaining extremal ray is contained in $\overline{\text{NE}}(X)_{K_X \geq 0}$. The convexity fact above now gives

$$\overline{\text{NE}}(X) = \overline{\text{NE}}(X)_{K_X \geq 0} + \sum_i \mathbb{R}_{\geq 0}[C_i],$$

as required. In particular, each K_X -negative extremal ray is generated by a rational curve of anticanonical degree at most $\dim X + 1$. $\square$

Example 2.4.2 (Fano varieties). Let X be a Fano variety. Then $\overline{\text{NE}}(X) \subseteq N_1(X)_{\mathbb{R}}$ is a rational polyhedral cone, spanned by the classes of finitely many rational curves. Apply part (2) of the cone theorem with $H = -K_X$ and $\varepsilon = \frac{1}{2}$. Since $K_X + \varepsilon H = \frac{1}{2}K_X$ is negative on $\overline{\text{NE}}(X) - \{0\}$, the nonnegative part is $\{0\}$.

Now we introduce a key step: suitable extremal rays correspond to morphisms of a variety. This connects the cone of curves with more traditional geometric properties of the variety.

Definition 2.4.3. Let X be a normal projective variety and $F \subseteq \overline{\text{NE}}(X)$ an extremal face. A surjective morphism $\text{cont}_F : X \rightarrow Z$ to a normal projective variety is called the *contraction of F* if the following hold:

- (1) $\text{cont}_F(C)$ is a point for an irreducible curve $C \subseteq X$ if and only if $[C] \in F$, and F is the closed cone generated by these curve classes;
- (2) $(\text{cont}_F)_* \mathcal{O}_X = \mathcal{O}_Z$; in particular, cont_F has connected fibres.

Remark 2.4.4. By Proposition 1.4.38, the contraction of F is unique up to isomorphism if it exists. Moreover, a morphism $g : X \rightarrow Z$ between normal projective varieties with $g_* \mathcal{O}_X = \mathcal{O}_Z$ is determined by the curves it contracts. Its relative cone $\text{NE}(g)$, as before, is generated by their classes; its closure is the corresponding extremal face of $\overline{\text{NE}}(X)$. Thus g is the contraction of this face. Now we give a lemma which will be used in Chapter 5.

Lemma 2.4.5. *Let X be a normal projective variety and let $r \subseteq \overline{\text{NE}}(X)$ be an extremal ray. Assume that X is $\mathbb{Q}$ -factorial and that the contraction of r between normal projective varieties exists,*

$$f = \text{cont}_r : X \longrightarrow Z.$$

Then one of the following holds:

- *f is birational and $\text{Ex}(f)$ is an irreducible divisor (we say that f is a **divisorial contraction**);*
- *f is birational and $\text{Ex}(f)$ has codimension at least two (we say that f is a **small contraction**);*
- *$\dim Z < \dim X$ (we say that f is **of fibre type**).*

Proof. If $\dim Z = \dim X$, the equality $f_*\mathcal{O}_X = \mathcal{O}_Z$ implies that f is birational. Thus the only assertion to prove is that $\text{Ex}(f)$ is irreducible when it contains a divisor.

Assume that $\text{Ex}(f)$ contains a prime divisor D . Since Z is normal, $f(D)$ has codimension at least two in Z . After replacing a projective embedding of Z by a sufficiently high Veronese embedding, choose a general integral hyperplane section H containing $f(D)$. Equivalently, choose a general member of a sufficiently positive linear system with base locus $f(D)$.

Let $H' \subseteq X$ be the strict transform of H . The proper morphism $H' \rightarrow H$ is surjective, so H' meets $f^{-1}(z)$ for every $z \in f(D)$. On the other hand, H' does not contain D . Thus for general $z \in f(D)$, the fibre $f^{-1}(z)$ meets H' but is not contained in it.

This fibre is connected. Since X is projective, we can choose an irreducible curve $C \subseteq f^{-1}(z)$ which meets H' but is not contained in it: a component of the fibre outside H' meets H' , and hyperplane sections through an intersection point give such a curve. By $\mathbb{Q}$ -factoriality, H' is $\mathbb{Q}$ -Cartier, so $(H' \cdot C) > 0$. Since C is contracted by f , we have $r = \mathbb{R}_{\geq 0}[C]$.

The Cartier divisor H has pullback

$$f^*H = H' + \sum_i a_i E_i,$$

where the E_i are the exceptional prime divisors that occur in the pullback, and $a_i > 0$. Locally these coefficients are the orders of vanishing of a local equation of H pulled back to X . Each E_i is $\mathbb{Q}$ -Cartier. Moreover, $(f^*H \cdot C) = (H \cdot f_*C) = 0$, hence

$$-(H' \cdot C) = \left(\sum_i a_i E_i \cdot C \right) < 0.$$

It follows that there is a divisor $E_i = E' \subseteq \text{Ex}(f)$ with $(E' \cdot C) < 0$.

For a proper birational morphism to a normal variety, every point of $\text{Ex}(f)$ lies on a positive-dimensional fibre, by Zariski's main theorem. The fibres are connected, so through any $x \in \text{Ex}(f)$ there is an irreducible contracted curve C' . Since f contracts the ray r ,

$$[C'] = \lambda[C] \quad \text{for some } \lambda \in \mathbb{R}_{>0}.$$

Therefore $(E' \cdot C') = \lambda(E' \cdot C) < 0$. An effective $\mathbb{Q}$ -Cartier divisor has nonnegative intersection with a curve not contained in its support, so $C' \subseteq E'$. Hence $x \in E'$. We conclude that $\text{Ex}(f) = E'$, as wanted. $\square$

In general, an extremal face need not admit a contraction to a projective variety, as the following example shows.

Example 2.4.6. For this example, work over $k = \mathbb{C}$. Let $g : X \rightarrow \mathbb{P}^2$ be the blow-up at twelve distinct very general points $p_1, \dots, p_{12}$ on a smooth plane cubic D . Let $C \subseteq X$ be the strict transform of D . By the canonical divisor formula for a blow-up [Har77, II, Exercise 8.5(b)],

$$\begin{aligned} (K_X \cdot C) &= \left(g^* K_{\mathbb{P}^2} + \sum_{i=1}^{12} E_i \right) \cdot C \\ &= \deg \mathcal{O}_D(-3) + \sum_{i=1}^{12} (E_i \cdot C) \\ &= -3 \times 3 + 12 = 3. \end{aligned}$$

Since $g^*D = C + \sum_{i=1}^{12} E_i$, we also have

$$C^2 = \left(g^*D - \sum_{i=1}^{12} E_i \right)^2 = 9 - 12 = -3.$$

Thus, by Example 1.4.31, $[C]$ spans an extremal ray r of $\overline{\text{NE}}(X)$. We claim that r has no contraction to a projective variety.

Indeed, suppose that such a contraction $\text{cont}_r : X \rightarrow Z$ exists. Since $r \neq \overline{\text{NE}}(X)$, the variety Z is not a point. Pulling back an ample line bundle on Z , we obtain a nontrivial line bundle on X

$$L = g^* \mathcal{O}_{\mathbb{P}^2}(d) \otimes \mathcal{O}_X \left(\sum_{i=1}^{12} a_i E_i \right)$$

whose restriction to C is trivial. Identifying C with D , this gives

$$\mathcal{O}_D(d) \otimes \mathcal{O}_D \left(\sum_{i=1}^{12} a_i p_i \right) \simeq \mathcal{O}_D.$$

Its degree is $3d + \sum_i a_i$, so a relation is possible only if $3d + \sum_i a_i = 0$. For every nonzero integer tuple satisfying this degree condition, at least one a_i is nonzero. Varying that p_i , multiplication by a_i on the elliptic curve is nonconstant. Thus the displayed relation holds on a proper closed subset of D^{12} . There are only countably many integer tuples. Choosing the points very generally avoids all these subsets, so the only possible relation has $d = a_1 = \cdots = a_{12} = 0$. This contradicts the choice of L . Hence contr_r does not exist, as claimed.

2.4B Application: Minimal Models of Surfaces

For the rest of this section, all surfaces are smooth and projective over k . On a smooth surface, curves are Cartier divisors, and intersection identifies $N^1(X)_{\mathbb{R}}$ with $N_1(X)_{\mathbb{R}}$. We use this identification when writing z^2 for $z \in N_1(X)_{\mathbb{R}}$. To begin with, we need two lemmas.

Lemma 2.4.7. *If D is a Cartier divisor on a smooth projective surface X with $D^2 > 0$, then either $|nD| \neq \emptyset$ for all $n \gg 0$, or $|-nD| \neq \emptyset$ for all $n \gg 0$.*

Proof. By Riemann–Roch on a surface and Serre duality, dropping the nonnegative superabundance terms h^1 , we get

$$\begin{aligned} h^0(nD) + h^0(K_X - nD) &\geq \frac{n^2}{2}D^2 - \frac{n}{2}(D \cdot K_X) + \chi(\mathcal{O}_X), \\ h^0(-nD) + h^0(K_X + nD) &\geq \frac{n^2}{2}D^2 + \frac{n}{2}(D \cdot K_X) + \chi(\mathcal{O}_X). \end{aligned}$$

The right-hand sides grow quadratically. Set $B = h^0(2K_X)$. If both $H^0(K_X - nD)$ and $H^0(K_X + nD)$ are nonzero, multiplication by a fixed nonzero section of either line bundle injects the other space into $H^0(2K_X)$, since X is integral. Thus both dimensions are at most B . If one space is zero, the corresponding inequality already gives sections of nD or $-nD$.

For n sufficiently large that both right-hand sides exceed B , it follows that $h^0(nD) > 0$ or $h^0(-nD) > 0$. These signs cannot vary with n : for an ample divisor H , a nonzero effective divisor has positive intersection with H , whereas $(nD \cdot H)$ and $(-nD \cdot H)$ have opposite signs. This also shows $(D \cdot H) \neq 0$. Hence one fixed sign works for all $n \gg 0$. $\square$

Corollary 2.4.8. *Let X be a smooth projective surface with an ample Cartier divisor H . The set*

$$Q := \{z \in N_1(X)_{\mathbb{R}} \mid z^2 > 0\}$$

has two connected components,

$$Q^+ := \{z \in Q \mid (z \cdot H) > 0\}, \quad Q^- := \{z \in Q \mid (z \cdot H) < 0\}.$$

Furthermore, $Q^+ \subseteq \overline{\text{NE}}(X)$.

Proof. We have the intersection form

$$(-, -) : N_1(X)_{\mathbb{R}} \times N_1(X)_{\mathbb{R}} \longrightarrow \mathbb{R}, \quad (z_1, z_2) \longmapsto (z_1 \cdot z_2).$$

Write $\rho = \rho(X)$. By the Hodge index theorem,

$$N_1(X)_{\mathbb{R}} = \mathbb{R}[H] \oplus H^{\perp}, \quad H^{\perp} = \{z \in N_1(X)_{\mathbb{R}} \mid (z \cdot H) = 0\},$$

where the intersection form is positive definite on the one-dimensional space $\mathbb{R}[H]$ and negative definite on the $(\rho - 1)$ -dimensional space $H^{\perp}$.

More precisely, for a basis $\{D_i\}_{i=1}^{\rho}$, let $M = (D_i \cdot D_j)_{i,j}$. Writing $z_1 = \sum_i x_i D_i$ and $z_2 = \sum_i y_i D_i$, we have

$$(z_1 \cdot z_2) = x^T M y.$$

The real symmetric matrix M can be orthogonally diagonalized. After also rescaling the basis vectors, Sylvester's law of inertia gives a basis $\{e_i\}_{i=1}^{\rho}$, orthogonal for the intersection form, in which the matrix is

$$\begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & -1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & -1 \end{pmatrix}.$$

We may take $e_1 = [H]/\sqrt{H^2}$ and choose the remaining basis vectors in $H^{\perp}$, so

$$[H] = (\sqrt{H^2}, 0, \dots, 0).$$

For $z = \sum_i x_i e_i \in Q$, we have

$$x_1^2 - \sum_{i \geq 2} x_i^2 > 0, \quad \text{or equivalently} \quad |x_1| > \sqrt{\sum_{i \geq 2} x_i^2}.$$

This gives the two open convex, hence connected, components

$$Q^+ = \left\{ x_1 > \sqrt{\sum_{i \geq 2} x_i^2} \right\}, \quad Q^- = \left\{ x_1 < -\sqrt{\sum_{i \geq 2} x_i^2} \right\}.$$

Since $(z \cdot H) = x_1 \sqrt{H^2}$, these are exactly the components in the statement.

For the second assertion, first take a rational class $[D] \in Q^+$. After clearing denominators, we may assume that D is Cartier. By Lemma 2.4.7, either nD or $-nD$ is linearly equivalent to an effective divisor for $n \gg 0$. An effective nonzero divisor has positive intersection with H , so $(D \cdot H) > 0$ excludes the negative sign. Hence $[D] = \frac{1}{n}[nD] \in \overline{\text{NE}}(X)$. Rational classes are dense in the open set Q^+ , and $\overline{\text{NE}}(X)$ is closed. Thus $Q^+ \subseteq \overline{\text{NE}}(X)$. In particular, every point of Q^+ is an interior point of $\overline{\text{NE}}(X)$. $\square$

Now we give the main result in this section (even in this chapter!), classifying the possible contractions on surfaces when K_X fails to be nef.

Theorem 2.4.9 (MMP for Surfaces). *Let X be a smooth projective surface and $R \subseteq \overline{\text{NE}}(X)$ a K_X -negative extremal ray. Then the contraction morphism $\text{cont}_R : X \rightarrow Z$ exists and is of one of the following types:*

- (1) Z is a smooth projective surface and X is obtained from Z by blowing up a closed point; $\rho(Z) = \rho(X) - 1$.
- (2) Z is a smooth projective curve and X is a geometrically ruled surface over Z , that is, a $\mathbb{P}^1$ -bundle; $\rho(X) = 2$.
- (3) Z is a point, $\rho(X) = 1$ and $-K_X$ is ample. (In fact, $X \simeq \mathbb{P}^2$, but this requires an additional argument.)

Proof. Step 1: By the cone theorem, Theorem 2.4.1 and its proof, we may choose an irreducible rational curve $C \subseteq X$ with $R = \mathbb{R}_{\geq 0}[C]$ and

$$0 < -(K_X \cdot C) \leq \dim X + 1 = 3.$$

Hence $(K_X \cdot C) \in \{-1, -2, -3\}$. Recall that the arithmetic genus is defined by

$$p_a(C) := 1 - \chi(C, \mathcal{O}_C) = h^1(C, \mathcal{O}_C),$$

since C is integral and proper. As a consequence of Riemann–Roch for surfaces, we have the adjunction formula [Har77, V, Exercise 1.3(a)]:

$$2p_a(C) - 2 = (C \cdot (C + K_X)) = C^2 + (K_X \cdot C).$$

Since $p_a(C) \geq 0$ (notice that C may be singular!), this implies $C^2 \geq -2 - (K_X \cdot C) \geq -1$. Equality $C^2 = -1$ holds if and only if $(K_X \cdot C) = -1$ and $p_a(C) = 0$. In that case $C \simeq \mathbb{P}^1$, so C is a (-1) -curve. Castelnuovo’s contraction theorem contracts exactly C to a smooth point on a smooth projective surface Z . This is the contraction of R , and $\rho(Z) = \rho(X) - 1$, giving case (1).

Step 2: It remains to deal with $C^2 > 0$ and $C^2 = 0$. Suppose first that $C^2 > 0$. By Corollary 2.4.8, $[C] \in Q^+$ is an interior point of $\overline{\text{NE}}(X)$. Since $[C]$ also spans an extremal ray, this forces $\rho(X) = 1$ and $\overline{\text{NE}}(X) = R$. Thus $X \rightarrow \text{Spec } k$ is the contraction of R . By assumption $(K_X \cdot C) < 0$, so $-K_X$ is positive on $\overline{\text{NE}}(X) - \{0\}$ and is ample by Kleiman’s criterion, Theorem 1.4.28. For the further conclusion $X \simeq \mathbb{P}^2$, see the classification of smooth del Pezzo surfaces, or [Mat02, I-4-4].

Step 3: Now consider $C^2 = 0$. Adjunction gives $2p_a(C) - 2 = (K_X \cdot C) < 0$, so $p_a(C) = 0$, $(K_X \cdot C) = -2$ and $C \simeq \mathbb{P}^1$.

We prove that a multiple of C gives the desired contraction. For a fixed ample divisor H , $(K_X - mC) \cdot H < 0$ for $m \gg 0$, since $C \cdot H > 0$. Thus, by Serre duality,

$$H^2(X, \mathcal{O}_X(mC)) \simeq H^0(X, \mathcal{O}_X(K_X - mC))^\vee = 0 \quad (m \gg 0).$$

Riemann–Roch on the surface now gives

$$h^0(X, \mathcal{O}_X(mC)) \geq \chi(X, \mathcal{O}_X(mC)) = -\frac{m}{2}(K_X \cdot C) + \chi(\mathcal{O}_X) = m + \chi(\mathcal{O}_X),$$

which grows linearly. In particular, for m sufficiently large, $h^0(X, \mathcal{O}_X(mC)) \geq 2$.

Write the complete linear system as $|mC| = F + |M|$, where F is its fixed part. Since mC itself is a member of $|mC|$, we have $F \leq mC$ as divisors. Because C is irreducible, this implies $F = kC$ for an integer $0 \leq k < m$. Removing the fixed part gives the complete linear system

$$|M| = |(m - k)C|.$$

Set $m' = m - k > 0$. Then $|m'C|$ has no fixed component and has dimension at least one.

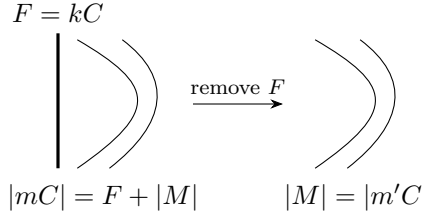


Figure 2.10: Removing the fixed component from $|mC|$ (schematic).

Next we verify that $|m'C|$ is base point free. Since it has no fixed component, we may choose $D \in |m'C|$ which does not contain C . Then

$$(D \cdot C) = m'C^2 = 0,$$

so D is disjoint from C . But $m'C$ is another member of the same system. These two members have disjoint supports, and hence $|m'C|$ has no base points.

Let $\phi : X \rightarrow Y \subseteq \mathbb{P}_k^N$ be the corresponding morphism. Its image has dimension one: it is nonconstant, and a two-dimensional image would imply $(m'C)^2 > 0$. Take the Stein factorization

$$\phi : X \xrightarrow{f} Z \xrightarrow{h} Y \subseteq \mathbb{P}_k^N, \quad Z = \text{Spec}_Y(\phi_* \mathcal{O}_X).$$

Here $f_* \mathcal{O}_X = \mathcal{O}_Z$, h is finite, and Z is a normal projective curve, hence smooth. We have not yet assumed that $f = \text{cont}_R$.

Put $A = h^* \mathcal{O}_Y(1)$, so $f^* A \simeq \mathcal{O}_X(m'C)$. By the projection formula and $f_* \mathcal{O}_X = \mathcal{O}_Z$, the section with zero divisor $m'C$ is the pullback of a section of A . Its zero divisor on Z is supported at one point z_0 , since its pullback is supported on the irreducible curve C . Hence the scheme-theoretic fibre over z_0 is aC for some integer $a > 0$.

The morphism f is flat, since it is a dominant morphism from a smooth surface to a smooth curve. All its fibres are numerically equivalent as divisors. A general fibre G is smooth and connected by generic smoothness in characteristic zero and the Stein factorization. Thus $G \equiv aC$, and adjunction gives

$$2g(G) - 2 = G^2 + (K_X \cdot G) = a(K_X \cdot C) = -2a.$$

Since $g(G) \geq 0$ and $a > 0$, we get $a = 1$ and $g(G) = 0$. Therefore every fibre has numerical class $[C]$.

Let $\sum_i a_i C_i$ be any fibre, with $a_i \in \mathbb{Z}_{>0}$ and C_i its distinct irreducible components. We now have

$$\sum_i a_i [C_i] = [C] \in R.$$

Since R is extremal, every $[C_i]$ lies in R . Consequently $C_i^2 = 0$ and $(K_X \cdot C_i) < 0$. Adjunction gives

$$2p_a(C_i) - 2 = (K_X \cdot C_i) < 0,$$

so $p_a(C_i) = 0$, $C_i \simeq \mathbb{P}^1$ and $(K_X \cdot C_i) = -2$. It follows that

$$-2 = (K_X \cdot C) = \left(K_X \cdot \sum_i a_i C_i \right) = -2 \sum_i a_i.$$

Thus $\sum_i a_i = 1$, and every fibre is irreducible, reduced, and isomorphic to $\mathbb{P}^1$.

Moreover, an irreducible curve E is contracted by f if and only if $[E] \in R$. One direction follows from the description of the fibres. For the other, if $[E] \in R$, then $(f^*A \cdot E) = m'(C \cdot E) = 0$; since A is ample on Z , E must be contracted. Hence $f = \text{cont}_R$.

Finally, Tsen's theorem gives a $k(Z)$ -rational point on the genus-zero generic fibre. This yields a section over an open subset of Z , which extends over all of the smooth curve Z by properness. Thus X is a geometrically ruled surface over Z . By [Har77, V, Proposition 2.2], it is the projectivization of a rank-two locally free sheaf on Z :

$$X \simeq \mathbb{P}_Z(\mathcal{E}).$$

In particular, $\rho(X) = \rho(Z) + 1 = 2$, as also follows from [Har77, II, Exercise 7.9(a)]. This is case (2). $\square$

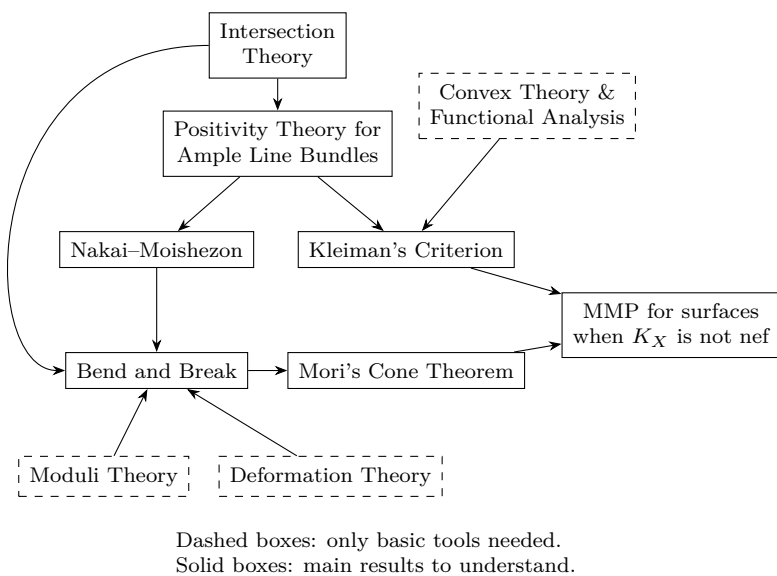
Summary:

Figure 2.11: Knowledge framework.

References for this excerpt

References to numbered statements outside Section 2.4 are to the original notes. The following short list identifies the external references used here; it is not an additional part of Section 2.4.

Har77

Robin Hartshorne, *Algebraic Geometry*, Graduate Texts in Mathematics 52, Springer, 1977.

Mat02

Kenji Matsuki, *Introduction to the Mori Program*, Universitext, Springer, 2002.

Debarre

Olivier Debarre, *Introduction to Mori Theory*, 2010–2011, especially the cone theorem. Author’s lecture notes.

Convexity

Olivier Debarre, *Bend and Break*, Grenoble lecture notes, 2007, Section 9.1, Lemma 21(b). Author’s lecture notes.

Tsen The Stacks Project, Theorem 59.67.10, Tag 03RD. Tsen’s theorem.